



Mansoura University
Faculty of Computers and Information
Department of Computer Science



Spring 2026

[SWE143] Software Project Management

Lecture 05: **Project Initiation**

Level 300: software engineering program

Muhammad Haggag Zayyan, Ph.D

Project Initiation: Feasibility and Project Authorization

Initiating a project

Project Initiation is the first phase of a project where you decide whether the project should start and get official approval.

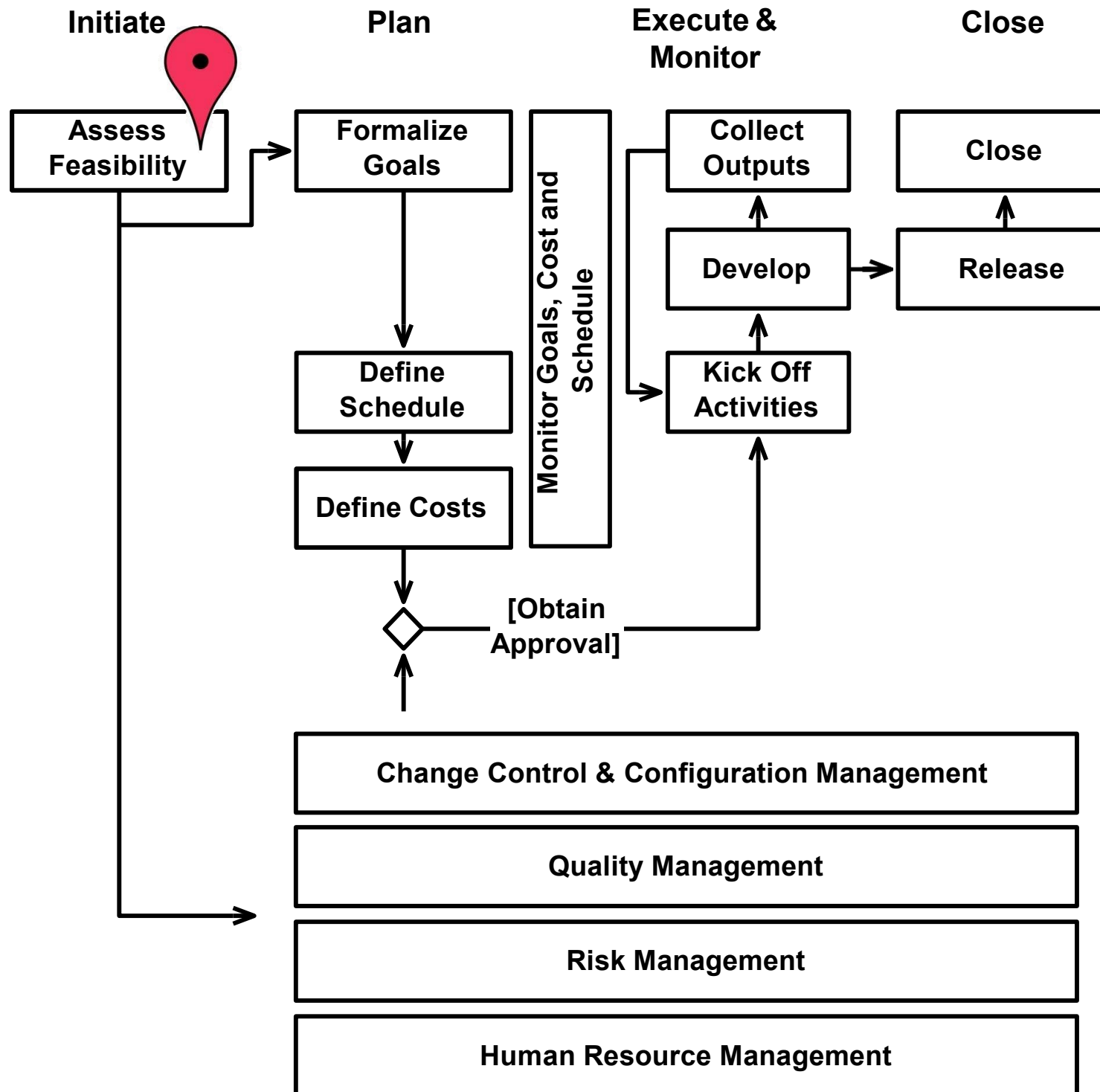
- **Feasibility:** **Can we do it?** → Checking if the project is possible, practical, and worth the time and cost.
- **Project Authorization:** **Can we start it?** → Giving official approval to begin the project and allocate resources.

Goals of this Unit

- Learning qualitative نوعی and quantitative کمی techniques to select among different projects
- Learning qualitative and quantitative techniques to choose the best alternative among different implementations of the same project
- Understanding how to write a Feasibility Study
- Choosing between internal development or external development (make or buy)

How does a project start?

- Initiation by some stakeholder (a company, a potential customer, ...) driven by a need (market, social, legal, technological advance, ...)
- Boundaries and process not always clear or very formalized
- First activities performed to:
 - Agree on the goals (scope)
 - Understand value and risks (for the performing organization and for the other stakeholders)
 - Choose a project approach



Make or Buy : How to select a project to invest?

internal vs. external development decision

- **Internal Development (Make)** – Developing the project **in-house** using your organization's resources, staff, and infrastructure.
- **External Development (Buy)** – Getting the project from outside the organization instead of building it internally. This can include:
 - ❑ **SaaS (Software as a Service)** → subscribe and use online (e.g., cloud apps)
 - ❑ **Off-the-shelf software** → buy and install locally (one-time license)
 - ❑ **Outsourced/custom-built** → external company builds it for you

Option	Pros	Cons
Internal Development (Make)	• Full control over project • Customized to exact needs • Better data security	• Higher cost • Takes more time • Requires skilled staff
External Development (Buy)	• Faster implementation • Lower initial cost • Access to expert vendors	• Less control • May not fully fit needs • Dependency on external provider

Project Value and Risks

Project Value and Risks

- Two main characteristics determine whether a project is worth starting:
 - The **value** (benefits) generated by the project [*e.g., Profit*]
 - The **risks** (uncertainties) associated to the project [*e.g., Timing, Budget*]
- An example a **make or buy** a software. Namely, choosing between developing a new system or acquiring an existing one with features similar to those needed.
- The meaning of value and risk depend upon many **factors**
 - ❑ Value and risks can be assessed **qualitatively** or **quantitatively**
- Sound evaluations are hard because projects and real-world conditions are uncertain and can change unexpectedly.
- Factors that contributes in assessing the value in a project, can be used with different purposes.
 - ❑ To decide whether a project is worth pursuing لا يستحق التنفيذ ام لا
 - ❑ To select which project to start from a portfolio (a collection of all proposed or ongoing projects in an organization) and prioritize them.
 - ❑ To choose the best project plan, given a project with different plans

Project Value and Risks - Factors

- **Project Value:** Why is the project useful?
 - **Direct and indirect value:** Profit from a new app + better customer satisfaction
 - **Sustainability:** A system that keeps working and useful for years
 - **Strategic alignment:** A project that supports company goals like digital transformation
- **Project Risks:** What could go wrong?
 - **Resource availability:** Not enough developers or budget
 - **Timing:** Project may be delayed
 - **Technical difficulties:** New technology may fail or be hard to implement
 - **Environment & constraints:** Legal rules or company restrictions may limit the project

Value: Direct and Indirect Value

- **Direct and Indirect Value** measures the positive and negative outcomes of a project and its outputs
- *Some metrics to consider include:*
 - **Revenues**, both direct and indirect

In software project management, an organization may develop software for internal use, external use.

- **Internal software** is built for employees to improve internal processes (e.g., HR or payroll systems).
 - **Direct value:** Cost savings, improved efficiency, faster processes, reduced errors
 - **Indirect value:** Better employee satisfaction, improved decision-making, organizational learning
- **External software** is developed for customers or the market to generate revenue (e.g., apps or SaaS products).
 - **Direct value:** Revenue from sales, subscriptions, licenses, or usage fees
 - **Indirect value:** Brand reputation, customer loyalty, market expansion, data insights

Value: Direct and Indirect Value (2)

- **Social and environmental impact** – does they affect the following:
 - **Environmental** impact: Energy use (e.g., data centers), electronic waste, carbon footprint, and resource consumption from building and running the software.
 - **Social** impact: Effects on people and society, including work, communication, access to services, culture, privacy, and accessibility.
- **Know-how acquired**
 - Skills, experience, and expertise gained by the team through developing and implementing the software project (e.g., technical skills, process improvement, domain knowledge).
- Direct and indirect value are strictly related to the business model and to the sustainability of the project outputs (see next slide)

Value: Sustainability

- **Sustainability** refers to the capacity of sustaining the project and its outputs after the project end.
- Taking into account the **operational costs** of a project's outputs and the way in which the project outputs will survive after a project end is an important consideration to understand whether a project is worth starting.
 - Often **overlooked** يتم تجاهلها , especially when project execution generates revenues

Value: Alignment with the Strategic Objectives

- Alignment with strategic objectives means **how closely a software project matches** and supports the organization's main goals. (such as: increasing revenue, improving efficiency, entering new markets, improving customer experience)
- When a project is well aligned, it directly helps the organization achieve its strategy and long-term success.
- Why alignment matters:
 - **Priority:** Well-aligned projects are given higher priority.
 - **Resources:** More budget, time, and staff are allocated to them.
 - **Support:** They receive stronger support from management and stakeholders.
- Examples of strategic alignment:
 - **Quality alignment:** The project improves product/service quality to increase customer satisfaction and brand reputation.
 - **Cost alignment:** The project reduces costs and improves efficiency to increase profitability.

Risks: Resource Availability

- Projects require the availability of human, financial, and technical resources within specific time frames.
 - Although it may be difficult to estimate all resources in advance, a quick feasibility or resource check is useful to ensure the project needs are realistic.
- Some aspects to consider include:
 - **Required resources:** What the project needs to be completed (people, budget, tools, technology).
 - **Current workload and availability:** How busy the team is now and whether they have time to take on new work.
 - **Future demand and capacity:** Expected future projects and whether the organization will still have enough resources later.
 - **Priority and importance:** How urgent and valuable the project is compared to other projects

Risks: Timing

- Many projects have **specific time-windows for the delivery of their outputs (production)**
- Deliver too early or too late and the outputs of the project might be useless
 - Launch a rocket to a planet [deterministic time]
 - For software development: Deliver software early is a goal of Agile development - allows for quick feedback from users and the ability to iterate and improve based on that feedback [Iterate fast, Deliver often]
 - “If you're not embarrassed by the first version of your product, you've launched too late.” - philosophy
- Consider, for instance, the race of competing firms in delivering similar products

Risks: Technical Difficulty and Uncertainty

- The success of many projects relies on the **actual capability of solving various technical challenges**, when the time comes
- Understanding what these challenges are is an important factor in determining the risks associated to a project
- Example:
 - In a mobile app project, the team plans to add a real-time chat feature. They anticipate possible technical issues with server load and connectivity. To manage this risk, they design a simpler messaging system as a backup.
 - When performance problems occur during testing, they use the simpler version to keep the project on schedule

Risks: Project environment and constraints

- Projects are influenced by various **constraints**, both **internal** and **external**.
- Stakeholders inside and outside the organization can influence the project, and regulations or standards may also limit what can be done.
- Examples:
 - Technology stack limitations (External):
 - Difficulties integrating legacy or outdated systems that restrict technical design and development.
 - Tight deadlines (Internal):
 - Short timelines set by the organization may lead to rushed development, affecting quality and features.
 - Budget constraints (Internal):
 - Limited funding within the organization may reduce testing, resources, or functionality, impacting overall quality.

Techniques to Assess Value and Risks

Techniques to Assess Value and Risks

- Different techniques are available to assess the value and risk generated by a project. Some are based on financial considerations (**quantitative**), while others are more **qualitative** (e.g., SWOT Analysis (Strengths, Weaknesses, Opportunities, Threats)).
- Financial Methods
 - Payback – duration
 - Return of Investment – cash
 - Net Present Value – cash
 - Internal Rate of Return – discount rate

Payback Period

The payback period فترة الاسترداد is the time taken to gain a financial return equal to the original investments

- Measured in months or years
 - An estimation of the project expenses and incomes determines the profits and losses.
- When using the payback period the projects/options that **minimize** the payback period are chosen in favor of the others
 - The payback is the year at which the project covers all expenses and starts earning. The shorter the payback, the better.

Simple Example

Payback Period = Initial Investment ÷ Cash Flow Per Year

Where:

- Initial Investment → Cash Outflow in Period 0
- Cash Flow Per Year → Annual Cash Flow Generated

Given the cost of opening up new store locations in different states, how long would it take for revenue from those new stores to pay back the entire amount of the investment?"

- If opening the new stores amounts to an initial investment of \$400,000 and the expected cash flows from the stores would be \$200,000 each year, then the period would be 2 years.

$$\text{Payback Period} = \$400\text{k} \div \$200\text{k} = 2 \text{ Years}$$

- So it would take two years before opening the new store locations has reached its break-even نقطة التعادل point and the initial investment has been recovered.

Example (different cash flow per year/expenses spread on more than one year)

	Project A		Project B		Project C	
Year 0	€	(50,000.00)	€	(20,000.00)	€	(15,000.00)
Year 1	€	30,000.00	€	(10,000.00)	€	15,000.00
Year 2	€	30,000.00	€	10,000.00	€	1,000.00
Year 3	€	1,000.00	€	60,000.00		
Year 4	€	1,000.00	€	50,000.00		
Expenses	€	(50,000.00)	€	(30,000.00)	€	(15,000.00)
Gains	€	62,000.00	€	120,000.00	€	16,000.00
Profit	€	12,000.00	€	90,000.00	€	1,000.00
Payback	2 years		3 years		1 year	

Remark: accounting style notation.

Negative numbers in **red - losses** and in parentheses

Discussion

- **Advantages**

- Simple, readily available data
- It reduces exposure to risk
- Particularly effective in high-technology/fashion projects: need rapid returns on investment due to fast changes in technology, or customer preferences.
- It favors shorter term benefits: need rapid cash flow

- **Disadvantages**

- Difficult to use on longer term projects
- Does not quantify exposure to risk
- Does not look at total gains: doesn't consider the overall profitability

Payback Weaknesses

- Different projects might have the same payback period, but different profiles in returning of the investments
 - These profiles are not taken into account by the technique but could make the difference between two projects
 - Two projects with the same payback period could have very different cash flow patterns. For example:
 - **Project A:** The initial investment is recovered quickly, but the cash inflows after the payback period are smaller.
 - **Project B:** The initial investment is recovered more slowly, but the cash inflows after the payback period are much larger.
- Check an example in the next slide

Payback Weaknesses

Same payback period = 3 Years. The method may select Project A or B, but **Project A gets more money first (and reduces risks)**

Disadvantage: Does not quantify exposure to risk

Year	Project A	Project B
Year 0	€ (10,000.00)	€ (10,000.00)
Year 1	€ (5,000.00)	€ (5,000.00)
Year 2	€ <u>10,000.00</u>	€ 5,000.00
Year 3	€ 5,000.00	€ 10,000.00

Payback Weaknesses

Different payback periods. The method select Project B as payback period = 3 Years, but gets less money than Project A

Disadvantage: Does not look at total gains

Year	Project A	Project B
Year 0	€ (10,000.00)	€ (10,000.00)
Year 1	€ (5,000.00)	€ (5,000.00)
Year 2	€ 5,000.00	€ 5,000.00
Year 3	€ 5,000.00	€ 11,000.00
Year 4	€ 20,000.00	

Return on Investment (ROI)

ROI عائد الاستثمار calculates the average annual profit and transforms it into a percentage of the total investments

Profit = Returns – Investments

Annual Profit = Profit / Duration

ROI = Annual Profit / Investments

- When using ROI, choose the project with the highest ROI

Example

Suppose we have the following projections for a project we need to decide whether to start or not

	Project A		Project B		Project C	
Year 0	€	(50,000.00)	€	(20,000.00)	€	(15,000.00)
Year 1	€	30,000.00	€	(10,000.00)	€	15,000.00
Year 2	€	30,000.00	€	10,000.00	€	1,000.00
Year 3	€	1,000.00	€	60,000.00		
Year 4	€	1,000.00	€	50,000.00		

Example

- **Project A**

- Profit = $62000 - 50000 = 12000$
- Annual Profit = $12000 / 4 = 3000$
- ROI = $3000 / 50000 = 6\%$

- **Project B**

- Profit = $120000 - 30000 = 90000$
- Annual Profit = $90000 / 4 = 22500$
- ROI = $22500 / 30000 = 75\%$

- **Project C**

- Profit = $16000 - 15000 = 1000$
- Annual Profit = $1000 / 2 = 500$
- ROI = $500 / 15000 = 3\%$

SOLUTION: Project B (highest ROI)

Discounted Cash Flows/Inflation

- The value of money decreases over the years (inflation التضخم) according to the inverse compound interests formula

$$\text{Discount Factor} = \frac{1}{(1 + r)^t}$$

Where,
 r is the inflation rate,
 t is time period

- Thus, giving it the money we invest **now** the same weight of money we will get in five year is over optimistic
- DCF (Discounted Cash Flows) are techniques that take into account inflation
- Curiosity: where does inflation comes from?
A nice reference: <http://en.wikipedia.org/wiki/Inflation>

Net Present Value

- Due to inflation issue, longer projects, payback and ROI tend to overestimate profits.
- Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time
- NPV **discounts** inflation amounts in the future in order to provide a more realistic comparison between presents investments and future gains.
- Net Profit and losses are computed by:

$$NPV = \sum_{t=1}^T \frac{C_t}{(1+r)^t} - C_0$$

C_t = net cash inflow during the period t
 C_0 = total initial investment costs
 r = discount rate
 t = number of time periods

$$\text{Value} = \frac{1}{(1+r)^t} * \text{amount}$$

where r is the inflation rate, *amount* is the net profit at year t
Note that the first project year is year 0

Net Present Value Example

Hypothesis Discount
Rate: 10% (this is “*r*”)

$$\text{Net Present Value (NPV)} = -C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_T}{(1+r)^T}$$

$-C_0$ = Initial Investment r = Discount Rate
 C = Cash Flow T = Time

Year (<i>t</i>)	Cash Flow	Discount Factor	Present Value
0	€ (35,000.00)	1.00	€ (35,000.00)
1	€ 10,000.00	0.91	€ 9,090.91
2	€ 15,000.00	0.83	€ 12,396.69
3	€ 20,000.00	0.75	€ 15,026.30
Expenditure	€ (35,000.00)		€ (35,000.00)
Gains	€ 45,000.00		€ 36,513.90
Profit	€ 10,000.00		€ 1,513.90

The rule is to accept any project that has a **positive** NPV because it adds value to the firm. If there are multiple projects, select the one with the **highest** NPV.

Net Present Value: Discussion

- Advantages

- More accurate profit-loss data: NPV accounts for the time value of money-The idea is that a dollar today is worth more than a dollar in the future. (*Time Value of Money (TVM) Concept*)

- Disadvantages

- It uses a fixed discount rate (may be unrealistic): where the market, inflation, or risk profile changes over time.
- It favors shorter terms projects: For projects with long payback periods, the future cash flows might be too heavily discounted, making them seem less valuable than they actually are.

Internal Rate of Return معدل العائد الداخلي

- The Internal Rate of Return (IRR) is the discount rate that makes the Net Present Value (NPV) of a project equal to zero. It represents the expected compound annual return العائد السنوي المركب from an investment.
- IRR is compared with the company's **hurdle** rate (minimum required return). If IRR is greater than or equal to the hurdle rate, the project is accepted.
- IRR is also used to compare and rank projects, with higher IRR values generally preferred as they indicate higher returns.
- However, IRR assumes that all future cash flows are reinvested at the IRR itself, which is often unrealistic since firms may reinvest at different rates.

Internal Rate of Return

The IRR formula is as follows:

Use Excel function
=IRR(...)

Previous Example

Discount ratio	10.00%	
Year (t)	Cash Flow	CF/(1+r)^t
0	(\$35,000)	(\$35,000)
1	\$10,000	\$9,090.91
2	\$15,000	\$12,396.69
3	\$20,000	\$15,026.30
	sum	\$1,513.90

Discount ratio	12.18149%	
Year (t)	Cash Flow	CF/(1+r)^t
0	(\$35,000)	(\$35,000)
1	\$10,000	\$8,914.24
2	\$15,000	\$11,919.56
3	\$20,000	\$14,167.19
	sum	\$0.00

IRR= 12.18149%

$$0 = CF_0 + \frac{CF_1}{(1 + IRR)} + \frac{CF_2}{(1 + IRR)^2} + \frac{CF_3}{(1 + IRR)^3} + \dots + \frac{CF_n}{(1 + IRR)^n}$$

Or

$$0 = NPV = \sum_{n=0}^N \frac{CF_n}{(1 + IRR)^n}$$

Where:

CF_0 = Initial Investment / Outlay

$CF_1, CF_2, CF_3 \dots CF_n$ = Cash flows

n = Each Period

N = Holding Period

NPV = Net Present Value

IRR = Internal Rate of Return

Discount ratio	12.19%	
Year (t)	Cash Flow	CF/(1+r)^t
0	(\$35,000)	(\$35,000)
1	\$10,000	\$8,913.45
2	\$15,000	\$11,917.44
3	\$20,000	\$14,163.40
	sum	(\$5.71)

Decision Example:

•Required Rate of Return: 10%

•IRR of Project: 12%

Since 12% > 10%, the project is considered profitable and worth investing in.

Example of the IRR

An investor made an investment of \$500 and got \$570 next year. Calculate the internal rate of return on the investment.

Solution:

Invested amount, $CF_0 = (-\$500)$ (negative, because money went out)

Cash inflow after 1 year, $CF_1 = \$570$

Using internal rate of return formula,

$$0 = CF_0 + \frac{CF_1}{(1 + IRR)^1}$$

$$0 = -\$500 + \frac{570}{(1 + IRR)^1}$$

$$500 + 500 \times IRR = 570$$

$$IRR = 70/500$$

$$IRR = 0.14 = 14\%$$

If the 14% internal rate of return exceeds the company's minimal acceptable rate of return, the project is economically attractive

IRR vs NPV

Year	Project "A"	Project "B"
0	-400	-500
1	325	325
2	200	325
IRR	22.17%	19.43%
NPV @ 10%	60.74	64.05

A and B are mutually exclusive. Which project should you accept?

- Based on IRR rule: $IRR_A > IRR_B$. Then select **project A**
- Based on NPV rule: $NPV_A < NPV_B$. Then select **project B**
- Whenever there is a **conflict** (each method suggests selecting a different project) between NPV and IRR rule, you should always use NPV, so select **project B**
 - **Scale Matters:** IRR **ignores** project size. A smaller project with a higher IRR may seem better but could add less value than a larger project with a lower IRR. NPV shows the true value added.

Which Approach Performs Better?

- **Internal Development (Make):** Building the project in-house using company staff and resources.
 - “Make” often has higher fixed cost but lower long-term variable cost
- **External Development (Buy):** Outsourcing the project to an external vendor or specialist.
 - “Buy” often has lower upfront cost but ongoing payments

Metric	Internal Development	External Development
Payback Period	May be longer due to setup and training time	May be shorter due to faster implementation
NPV <i>(Measures value in today's money)</i>	Can be higher if used long-term and efficiently managed	Can be lower or higher depending on vendor cost and contract
ROI <i>(Measures percentage return)</i>	Can be higher long-term if fixed costs are spread over time	Can be higher short-term due to lower initial investment

Decision-Making Tip:

When comparing internal vs. external development, use these metrics together to get a balanced view:

- **Calculate Payback Period** to assess how quickly the project recovers costs.
- **Evaluate NPV** to understand long-term financial benefits.
- **Analyze ROI** to ensure the project meets your profitability expectations.

Score Matrices

- The financial methods (Payback, ROI, NPV) look only at some of the financial data
- A score matrix is a list of project criteria:
 - ❑ They are based on a standardized set of criteria and weights, which highlight the relevant features of a project.
 - The criteria highlight the desirable (value –e.g. profit) and undesirable (risk –e.g. timing) aspects of a project.
 - ❑ Scoring matrices allow one to take into account other factors
- A **qualitative** evaluation assesses how well a project meets each criterion by placing it on a scale (e.g., high, medium, low). This approach helps compare the project with past or competing projects by providing a clearer picture of its strengths and weaknesses.
- Even though the **criteria** might originate from **qualitative judgments**, the matrix itself transforms them into **quantitative data** for decision-making.

Score Matrix Example

Factor	Value	Weight	SUM
The project aligns with the strategic objectives	YES	2	2
The project has a profit > 20%	NO	4	0
Payback period < 2 years	YES	5	5
Enlarges the customer base	YES	2	2
The project requires a standard technology	NO	3	0
The quality constraints are simple to meet	YES	1	1
The timing is not too tight	NO	4	0
We have skilled personnel to do the work	YES	5	5

15

- **Quantitative** criteria are numeric and measurable.
- **Qualitative** criteria are descriptive and subjective.

- **Value** can be binary (YES/NO) or a number (e.g. from 1 to 5) and measures how well the project meets the requirement
- The **weight** measures how important a factor is for the **decision**
- **Sum** = multiply the scores with the weights
- Projects scoring a **higher** value are more desirable than projects with a lower score.

Score Matrix Example

- ❑ The selection process will tend to favor projects in which **stakeholders are not difficult to manage.**
- ❑ According to the data, "**Project 1,**" the one with the highest score, is preferred over the others.

Factor	Description	Weight	Project 1		Project 2		Project 3	
			Value	Total	Value	Total	Value	Total
Profit >30%	The project will yield a profit >30% if no exceptional events occur	3	4	12	4	12	4	12
Low-risk profile	The project does not present particular risks. That is, there is no risk with a very high impact	2	2	4	3	6	3	6
Schedule is not tight	Project delivery does not require activities to be performed in a very tight schedule	3	3	9	2	6	2	6
Manageable complexity	The complexity is manageable	2	1	2	1	2	1	2
Consistent with current business	The project is mainstream with the activities of the organization	1	1	1	1	1	1	1
Stakeholders	Stakeholders are difficult to manage	-4	2	-8	4	-16	3	-12
			20		11		15	

Positive or negative weights in a Score Matrix depend on whether a criterion impacts the project positively or negatively.

Discussion

- **Advantages**

- Simple
- It encourages standardization and more objectivity in decision making
- It helps discuss and evaluate the project characteristics
- It widens the range of evaluation
- Not biased toward shorter term projects

- **Disadvantages**

- A simple model may encourage development of long and unhelpful criteria lists.
- Different factors have same importance (unless the weight matrix is used)

Caveat

- Not all score matrices are equally good.
- The following is an example of a bad matrix.

Why?

Factor	Value	Weight	SUM	Comment
The project has a profit > 20%	YES	3	3	
The project is highly risky	NO	3	0	
			3	

SOLUTION

A positive factor (first row) and a negative factor (second row) influence in the same way the matrix.
As a consequence an highly risky project is preferred over a project which is not very risky

Good matrix:

Make sure the questions either all positively influence, or all negatively influence the decision or use scores with different signs (e.g., -ve weight)

Thank You